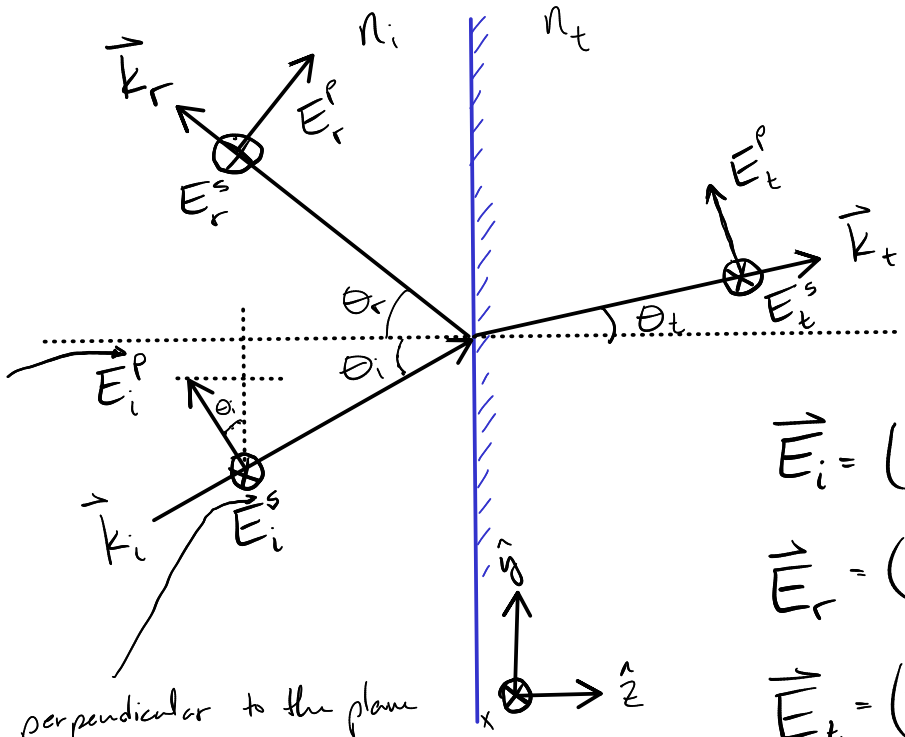
Chapter 3 - Reflection + Refraction

260316 M



$$\vec{k}_i = k_i(\hat{y}\sin\theta_i + \hat{z}\cos\theta_i)$$

$$\vec{k}_r = k_r(\hat{y}\sin\theta_r - \hat{z}\cos\theta_r)$$

$$\vec{k}_t = k_t(\hat{y}\sin\theta_t + \hat{z}\cos\theta_t)$$

"parallel" any vector in the plane is parallel to the plane

perpendicular to the plane "senkrecht"

$$\vec{E}_i = \left(E_i^p(\hat{y}\cos\theta_i - \hat{z}\sin\theta_i) + \hat{x}E_i^s\right)e^{i(k_i(y\sin\theta_i + z\cos\theta_i) - \omega_i t)}$$

$$\vec{E}_r = \left(E_r^p(\hat{y}\cos\theta_r + \hat{z}\sin\theta_r) + \hat{x}E_r^s\right)e^{i(k_r(y\sin\theta_r - z\cos\theta_r) - \omega_r t)}$$

$$\vec{E}_t = \left(E_t^p(\hat{y}\cos\theta_t - \hat{z}\sin\theta_t) + \hat{x}E_t^s\right)e^{i(k_t(y\sin\theta_t + z\cos\theta_t) - \omega_t t)}$$

from Maxwell's

↳ Boundary conditions require the components parallel to the interface to be continuous: ($\hat{x}$ + $\hat{y}$ directions in this example)

$$\left(E_i^p\,\hat{y}\cos\theta_i + \hat{x}E_i^s\right)e^{i(k_i(y\sin\theta_i + z\cos\theta_i) - \omega_i t)} + \left(E_r^p\,\hat{y}\cos\theta_r + \hat{x}E_r^s\right)e^{i(k_r(y\sin\theta_r - z\cos\theta_r) - \omega_r t)}$$

$$= \left(E_t^p\,\hat{y}\cos\theta_t + \hat{x}E_t^s\right)e^{i(k_t(y\sin\theta_t + z\cos\theta_t) - \omega_t t)}$$

$\omega_i = \omega_r = \omega_t = \omega$

$$\underbrace{k_i \sin\theta_i = k_r \sin\theta_r}$$ $= k_t \sin\theta_t$

$k = \frac{\omega}{v}$

$v = \frac{c}{n}$

$k_i = \frac{\omega}{v} = \frac{n_i \omega}{c}$ $\quad k_r = \frac{\omega}{v} = \frac{n_i \omega}{c}$

$k_i = k_r$

$\sin\theta_i = \sin\theta_r$

$\boxed{\theta_i = \theta_r}$ law of reflection

$k_i \sin\theta_i = k_t \sin\theta_t$

$\frac{n_i \cancel{\omega}}{\cancel{c}} \sin\theta_i = \frac{n_t \cancel{\omega}}{\cancel{c}} \cdot \sin\theta_t$

$\boxed{n_i \sin\theta_i = n_t \sin\theta_t}$ Snell's Law

$$E_i^p\,\hat{y}\cos\theta_i + \hat{x}E_i^s + E_r^p\,\hat{y}\cos\theta_r + \hat{x}E_r^s = E_t^p\,\hat{y}\cos\theta_t + \hat{x}E_t^s$$

x-direction

$$(1)\quad E_i^s + E_r^s = E_t^s$$

y-direction

$$E_i^p\cos\theta_i + E_r^p\cos\theta_r = E_t^p\cos\theta_t$$

$$\theta_i = \theta_r$$

$$(2)\quad \left(E_i^p + E_r^p\right)\cos\theta_i = E_t^p\cos\theta_t$$

So we have 2 equations but 4 unknowns, so we need the magnetic field equations.

$$\vec{B} = \frac{\vec{k}\times\vec{E}}{\omega} = \frac{n}{c}\,\hat{u}\times\vec{E} \quad \text{where} \quad \hat{u} = \frac{\vec{k}}{|k|}$$

$$(3)\quad n_i\left(E_i^s - E_r^s\right)\cos\theta_i = n_t E_t^s\cos\theta_t$$

$$(4)\quad n_i\left(E_i^p - E_r^p\right) = n_t E_t^p$$

Fresnel Equations

$$r_s = \frac{E_r^s}{E_i^s}$$

$$t_s = \frac{E_t^s}{E_i^s}$$

$$E_i^s + E_r^s = E_t^s \quad \Big| \quad \frac{\cancel{n_i \cos\theta_i}\left(E_i^s - E_r^s\right)}{\cancel{n_i \cos\theta_i}} = \frac{n_t \cos\theta_t \cdot E_t^s}{n_i \cos\theta_i}$$

solve w/ elimination

$$E_i^s - E_r^s = \frac{n_t \cos\theta_t \cdot E_t^s}{n_i \cos\theta_i}$$

$$2E_i^s = \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i} + 1\right) E_t^s$$

$$2E_i^s = \frac{n_t \cos\theta_t + n_i \cos\theta_i}{n_i \cos\theta_i} \cdot E_t^s$$

$$\boxed{t_s = \frac{E_t^s}{E_i^s} = \frac{2 n_i \cos\theta_i}{n_t \cos\theta_t + n_i \cos\theta_i}}$$

$$\boxed{t_p = \frac{E_t^p}{E_i^p} = \frac{2 n_i \cos\theta_i}{n_i \cos\theta_t + n_t \cos\theta_i}}$$

$$\boxed{r_s = \frac{E_r^s}{E_i^s} = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t}}$$

$$\boxed{r_p = \frac{E_r^p}{E_i^p} = \frac{n_i \cos\theta_t - n_t \cos\theta_i}{n_i \cos\theta_t + n_t \cos\theta_i}}$$

recall:

$$n_i \sin\theta_i = n_t \sin\theta_t$$

$$\theta_t = \sin^{-1}\left(\frac{n_i}{n_t} \sin\theta_i\right)$$

Reflectance + Transmittance

260320 F

Conservation of energy: $P_i = P_r + P_t$ ← power

$$P_i^s = P_r^s + P_t^s$$

$$P_i^p = P_r^p + P_t^p$$

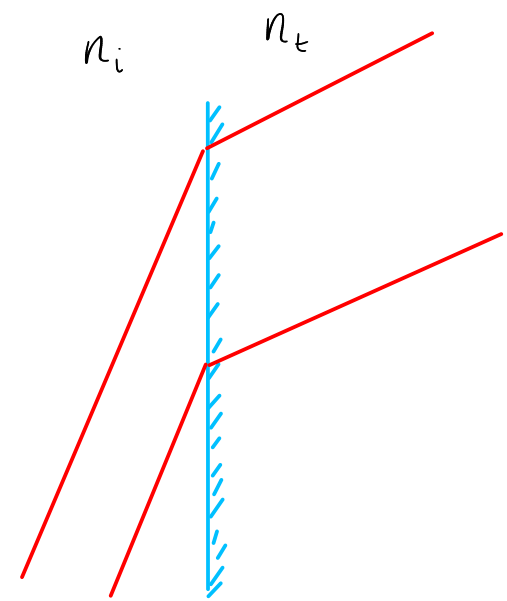
irradiance (intensity)

$$I = \frac{P}{A} \qquad I = c^2 \epsilon_0 \langle E^2 \rangle$$

$$P \propto I \propto E^2$$ ← electric field

reflectance

$$R_s = \frac{P_r^s}{P_i^s} = |r_s|^2 \qquad R_p = \frac{P_r^p}{P_i^p} = |r_p|^2$$



conservation of energy

$$R + T = 1 \Rightarrow \underline{\underline{T = 1 - R}}$$

$$T \neq t^2$$

$$T_s = \frac{n_t \cos\theta_t}{n_i \cos\theta_i} \cdot |t_s|^2 \qquad T_p = \frac{n_t \cos\theta_t}{n_i \cos\theta_i} \cdot |t_p|^2$$

Brewster's angle → the angle at which $r_p = 0$

$$\theta_i + \theta_t = \frac{\pi}{2}$$

Snell's → $n_i \sin\theta_i = n_t \sin\theta_t$

$$n_i \sin\theta_i = n_t \underbrace{\sin\left(\frac{\pi}{2} - \theta_i\right)}_{\cos\theta_i}$$ ← at one particular place

$$n_i \sin\theta_B = n_t \cos\theta_B$$ ← Brewster's angle

$$\tan\theta_B = \frac{n_t}{n_i} \Rightarrow \theta_B = \tan^{-1}\left(\frac{n_t}{n_i}\right)$$

Total Internal Reflection

Snell's law: $n_i \sin\theta_i = n_t \sin\theta_t$

solve for θ_t → $\theta_t = \sin^{-1}\left(\frac{n_i}{n_t} \cdot \sin\theta_i\right)$

as this grows

if $n_i / n_t > 1$

if arg $\sin^{-1} > 1$, $\sin^{-1}$ is complex

there is some angle where the light is no longer transmitted → it is reflected!

$$\theta_i \to \theta_c = \sin^{-1}\left(\frac{n_t}{n_i}\right) \qquad n_i > n_t$$

→ $R_s = 1$ $\quad R_p = 1$ $\quad \theta_i > \theta_c$

$$r_s = \frac{E_r^s}{E_i^s} = \frac{n_i\cos\theta_i - n_t\cos\theta_t}{n_i\cos\theta_i + n_t\cos\theta_t}$$

so we need a method for calculating $\cos\theta_t$

$$\cos^2\theta + \sin^2\theta = 1$$

$$\cos\theta_t = \sqrt{1-\sin^2\theta_t}$$

substitute using Snell's law

$$\cos\theta_t = \sqrt{1-\underbrace{\left(\frac{n_i}{n_t}\sin\theta_i\right)^2}_{\text{larger than 1 for total internal reflection}}}$$

$$r_s = \frac{n_i\cos\theta_i - i n_t\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}{n_i\cos\theta_i + i n_t\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}$$

$$r_p = -\frac{n_t\cos\theta_i - i n_i\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}{n_t\cos\theta_i + i n_i\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}$$

book writes:

$$\cos\theta_t = i\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}$$

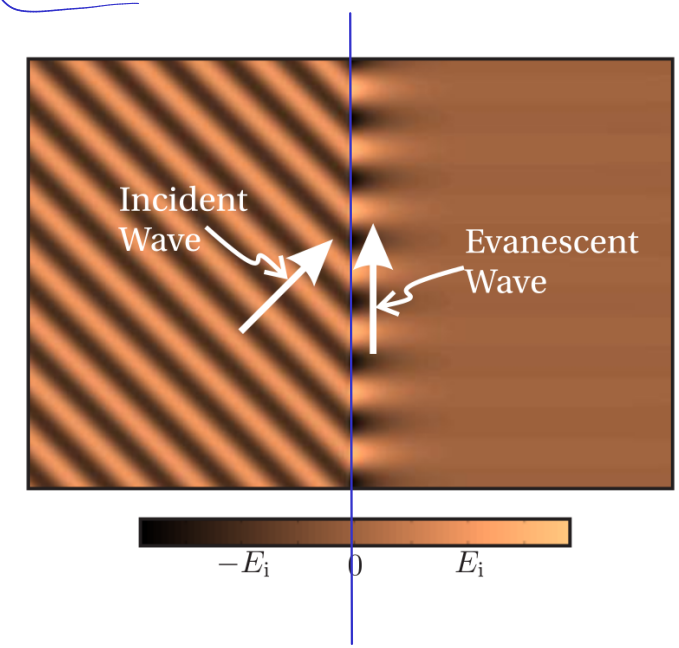
only for $n_i > n_t$

$$\vec{E}_t = \left(E_t^p\left(\hat{y}\cos\theta_t - \hat{z}\sin\theta_t\right) + \hat{x}E_t^s\right)e^{i(k_t(y\sin\theta_t + z\cos\theta_t) - \omega t)}$$

$$\vec{E}_t = \left(E_t^p\left(\hat{y}\,i\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1} - \hat{z}\frac{n_i}{n_t}\sin\theta_i\right) + \hat{x}E_t^s\right)e^{i\left(k_t\left(y\frac{n_i}{n_t}\sin\theta_i + iz\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}\right) - \omega t\right)}$$

$$\vec{E}_t = \left(E_t^p\left(\hat{y}\,i\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1} - \hat{z}\frac{n_i}{n_t}\sin\theta_i\right) + \hat{x}E_t^s\right)e^{-k_t z\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}\, e^{i\left(k_t y\frac{n_i}{n_t}\sin\theta_i - \omega t\right)}$$

$\underbrace{\qquad\qquad}_{\text{decaying in the } z\text{-direction}}$ $\underbrace{\qquad\qquad}_{\text{wave!}}$



Reflections from metal

$$n \rightarrow \tilde{n} = n + ik$$

$\downarrow$

textbook

$\mathcal{N}$

$$r_s = \frac{E_r^s}{E_i^s} = \frac{\tilde{n}_i \cos\theta_i - \tilde{n}_t \cos\theta_t}{\tilde{n}_i \cos\theta_i + \tilde{n}_t \cos\theta_t}$$

AND HW4

HW: 3.5, 7, 8, 10, 11, 12, 13, 14

15. When light passes through a prism its path changes once on the way into the prism and then again on the way out. Assume the prism is an equilateral triangle of glass ($n = 1.5$).

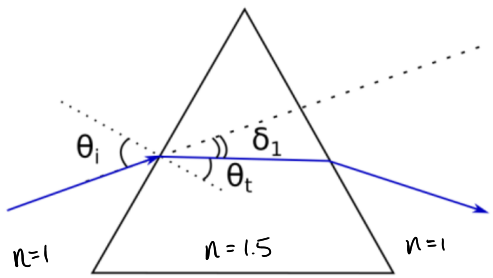
(a) For an exterior incident angle of 30°, what is the transmitted angle in the prism?

$$n_i \sin\theta_i = n_t \sin\theta_t$$

$$\sin\theta_t = \frac{n_i}{n_t}\sin\theta_i \Rightarrow \theta_t = \sin^{-1}\left(\frac{n_i}{n_t}\sin\theta_i\right)$$

$$\theta_t = 19.47°$$

(b) Inside the prism, what is the *deviation* δ_1 of the light's path?

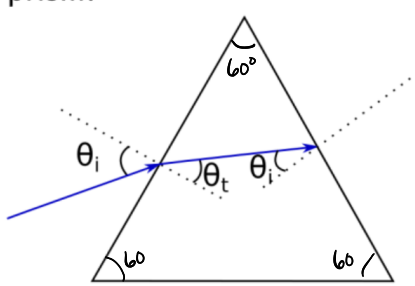


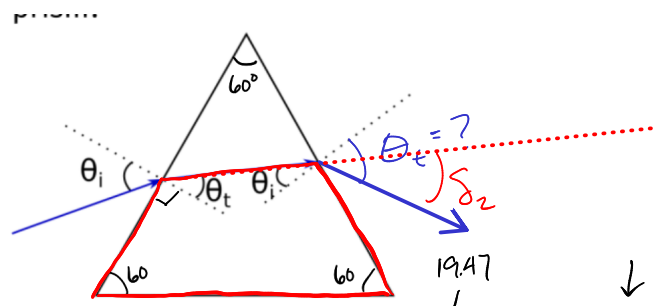
$$\theta_i = \delta_1 + \theta_t$$

$$\delta_1 = \theta_i - \theta_t$$

$$\delta_1 = 30° - 19.47° = 10.53°$$

(c) What is the interior incident angle as the light is leaving the prism?





$$90° - 19.47 = 70.53$$

$$180° - 60° - 70.53° = 49.47°$$

$$\theta_i = 90° - 49.47 = \underline{40.53°}$$

$$360 = 60 + 60 + (90 + \underbrace{\theta_t}_{19.47}) + (90 + \theta_i)$$

$$\theta_i = 40.53°$$

(d) What is the final transmitted angle as the light leaves the prism?

$n_i = 1.5$ $\quad n_t = 1$

$$\sin\theta_t = \frac{n_i}{n_t}\sin\theta_i$$

$$\theta_t = \sin^{-1}\left(\frac{1.5}{1}\sin(40.53)\right)$$

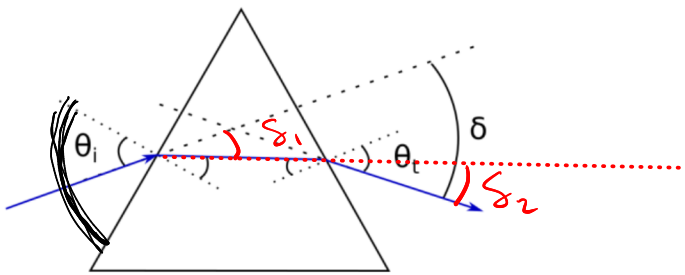
$$\theta_t = 77.1^\circ$$

(e) What is the second deviation of the light's path?

$$\delta_2 = \theta_t - \theta_i$$

$$= 77.1 - 40.53 = 36.57^\circ$$

(f) What is the total deviation of the lights path?



$$\delta = \delta_1 + \delta_2$$

$$= 10.53^\circ + 36.57^\circ$$

$$= 47.1^\circ$$

(g) Using excel or google sheets, do this procedure with many exterior incident angles (all integer angles). Don't forget that excel's trig functions only accept radians and inverse trig functions will return angles in radians so you will have to include some conversion factors in the formula (π rad $=$ 180°). Plot the total deviation as a function incident angle and see if there are any interesting features.

5

P3.5 A pentaprism is a five-sided reflecting prism used to deviate a beam of light by 90° without inverting an image (see Fig. 3.13). Pentaprisms are used in the viewfinders of SLR cameras.

(a) What prism angle β is required for a normal-incidence beam from the left to exit the bottom surface at normal incidence?

(b) If all interfaces of the pentaprism are uncoated glass with index $n = 1.5$, what fraction of the intensity would get through this system for a normal incidence beam? Compute for p-polarized light, and include transmission through the first and final surfaces as well as reflection at the two interior surfaces.

NOTE: You will find that the overall transmission through the device is very poor. The reflecting surfaces on pentaprisms are usually treated with a high-reflection coating and the transmitting surfaces are treated with anti-reflection coatings.

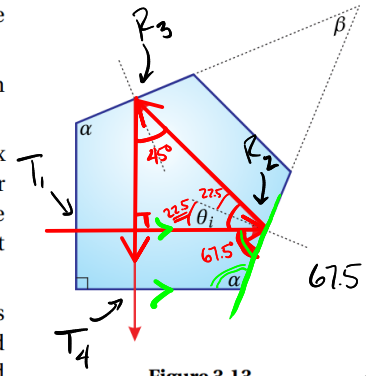


Figure 3.13

$67.5° + \alpha = 180$

$\alpha = 112.5°$

a) $90° + \alpha + \alpha + \beta = 360°$

$90 + 2(112.5) + \beta = 360°$

$\beta = 45°$

b) look at R + T

$T_1 \cdot R_2 \cdot R_3 \cdot T_4$

$\hookrightarrow T_p = 1 - R_p \; ; \; R_p = |r_p|^2 \; ; \; r_p = \dfrac{n_i \cos\theta_t - n_t \cos\theta_i}{n_i \cos\theta_t + n_t \cos\theta_i}$

$T_1 = 1 - R_1 = 1 - |r_{p1}|^2 = 1 - \left|\dfrac{1 - 1.5}{1 + 1.5}\right|^2$

$= 1 - |-0.2|^2$

$T_1 = 0.96$

7)

P3.7 (a) Find Brewster's angle for an air-glass interface with $n_{\text{glass}} = 1.5$.

(b) Compute R_s and R_p at this angle.

8)

P3.8 Diamonds have an index of refraction of $n = 2.42$ which allows total internal reflection to occur at relatively shallow angles of incidence. Gem cutters choose facet angles that ensure most of the light entering the top of the diamond will reflect back out to give the stone its expensive sparkle. One such cut, the "Eulitz Brilliant" cut, is shown in Fig. 3.14.

(a) What is the critical angle for diamond?

(b) What fraction of the light reflects for internal angles $\theta_i = 40.5°$ and $\theta_i = 50.6°$? One way to spot a fake diamond is by noticing reduced brilliance in the sparkle. Are these angles both beyond the critical angle for fused quartz ($n = 1.46$)?

(c) For each angle and assuming s-polarized light, find the phase shift upon reflection ϕ_s where $r_s = |r_s|\, e^{i\phi_s}$.

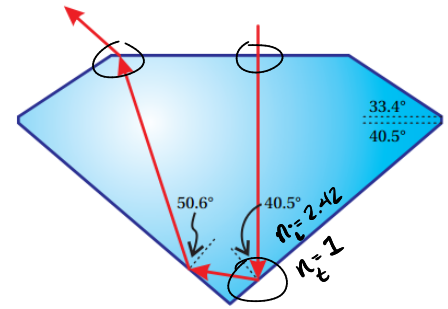


Figure 3.14 A Eulitz Brilliant cut diamond.

a) $\theta_c = \sin^{-1}\left(\frac{n_t}{n_i}\right)$

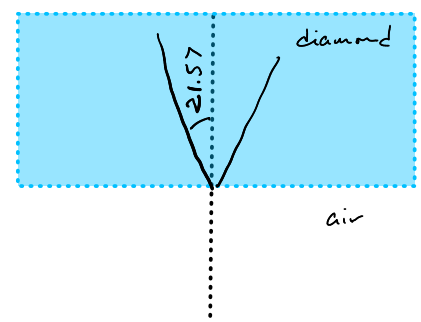
outside of the material

index of the material itself

$\theta_c = \sin^{-1}\left(\frac{1}{2.42}\right) = 21.57°$

$\theta_c = 43.2°$

fused quartz

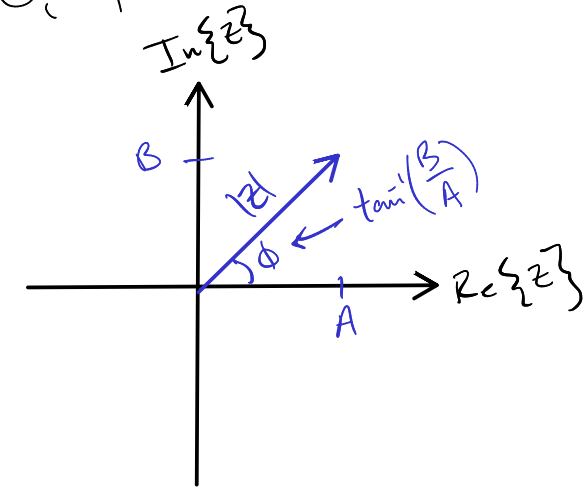


b) $R_{s1} = |r_{s1}|^2$

total internal reflection

$$r_s = \frac{n_i \cos\theta_i - i n_t \sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}{n_i \cos\theta_i + i n_t \sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}$$

$$r_p = -\frac{n_t \cos\theta_i - i n_i \sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}{n_t \cos\theta_i + i n_i \sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}$$

c) $r_s = |r_s| e^{i\phi_s}$ $\qquad r_s = A + iB$

$$|r_s| = \sqrt{r_s \cdot r_s^*} = \sqrt{A^2 + B^2}$$